neyman

STA258H5
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LAW OF LARGE NUMBERS

Importance: mathematically proves that large sample results are reliable

Review

Let RV $X \sim f(x)$, $E(x) = \mu$, $var(x) = \sigma^2$. $\forall k > 0$

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \iff P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

or

$$P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$$

A special case of Markov's

Convergence in Probability

The sequence of random variables $X_1,\ X_2,\ X_3, ..., X_n, ...$ is said to **converge in probability** to the constant c, if for every $\epsilon > 0$,

$$\lim_{n\to\infty} P\left(|X_n - c| \leq \epsilon\right) = 1$$

OR

$$\lim_{n\to\infty} P\left(|X_n - c| > \epsilon\right) = 0$$

Chebyshev's Inequality

If X is a random variable with finite mean μ and variance σ^2, then for any value $k > 0$,

$$P\left(|X - \mu| \geq k\right) \leq \frac{\sigma^2}{k^2}$$

Convergence in Probability

For a sequence of RV's $X_1, X_2, \ldots\ldots$ $\forall \varepsilon > 0$ the sequence converges to X if

$$\lim_{n \to \infty} P(|X_n - X| > \varepsilon) = 0$$

The Weak Law of Large Numbers (WLLN)

Let $X_1,\ X_2,\ \ldots$ be a sequence of independent and identically distributed random variables, each having finite mean $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Then, for any $\epsilon > 0$,

$$P\left[\left|\overbrace{\frac{X_1 + X_2 + \cdots + X_n}{n}}^{\bar{X}_n} - \mu\right| \geq \epsilon\right] \to 0 \quad \text{as} \quad n \to \infty$$

We write $\bar{X}_n \xrightarrow{\mathcal{P}} \mu$.

$$\lim_{n \to \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0$$

as n becomes larger, $\bar{X}_n$ gets closer to μ

Proof

Exercise

As
$E\left[\frac{X_1+X_2+\cdots+X_n}{n}\right] = \mu$ and $Var\left[\frac{X_1+X_2+\cdots+X_n}{n}\right] = \frac{\sigma^2}{n}$
it follows from Chebyshev's inequality that

$$P\left[\left|\frac{X_1+X_2+\cdots+X_n}{n} - \mu\right| \geq \epsilon\right] \leq \frac{\sigma^2}{n\epsilon^2}$$

and the result is proved.

Example

Let X_i for $i = 1, 2, 3, \ldots$ be independent Poisson random variables, with rate parameter $\lambda = 3$. Prove $\bar{X}_n \xrightarrow{\mathcal{P}} 3$.

Solution

Note that $E\left[\frac{X_1+X_2+\cdots+X_n}{n}\right] = 3$ and $Var\left[\frac{X_1+X_2+\cdots+X_n}{n}\right] = \frac{3}{n}$.
It follows from Chebyshev's inequality that for any $\epsilon > 0$,

$$P\left[\left|\frac{X_1+X_2+\cdots+X_n}{n} - 3\right| \geq \epsilon\right] \leq \frac{3}{n\epsilon^2}.$$

Therefore,

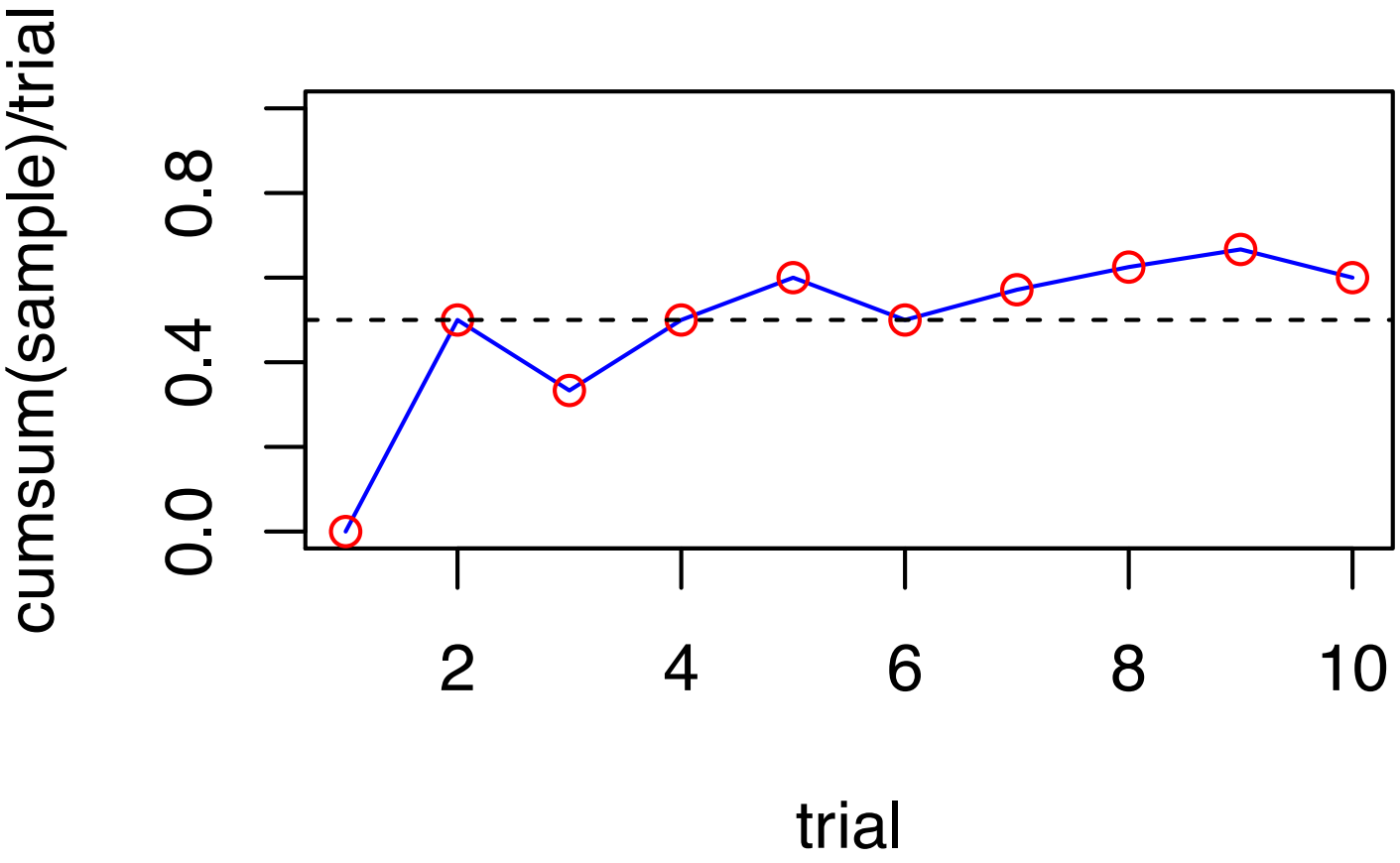
$$P\left[\left|\frac{X_1+X_2+\cdots+X_n}{n} - 3\right| \geq \epsilon\right] \to 0 \quad \text{as} \quad n \to \infty$$

and $\bar{X}_n \xrightarrow{\mathcal{P}} 3$.

Simulation

```
n=10;

trial=seq(1,n,by=1);

sample=rbinom(n,1,1/2);

plot(trial,cumsum(sample)/trial,type="l",ylim=c(0,1),
col="blue");

points(trial,cumsum(sample)/trial,col="red");

abline(h=0.5,lty=2, col="black");
```

Simulation

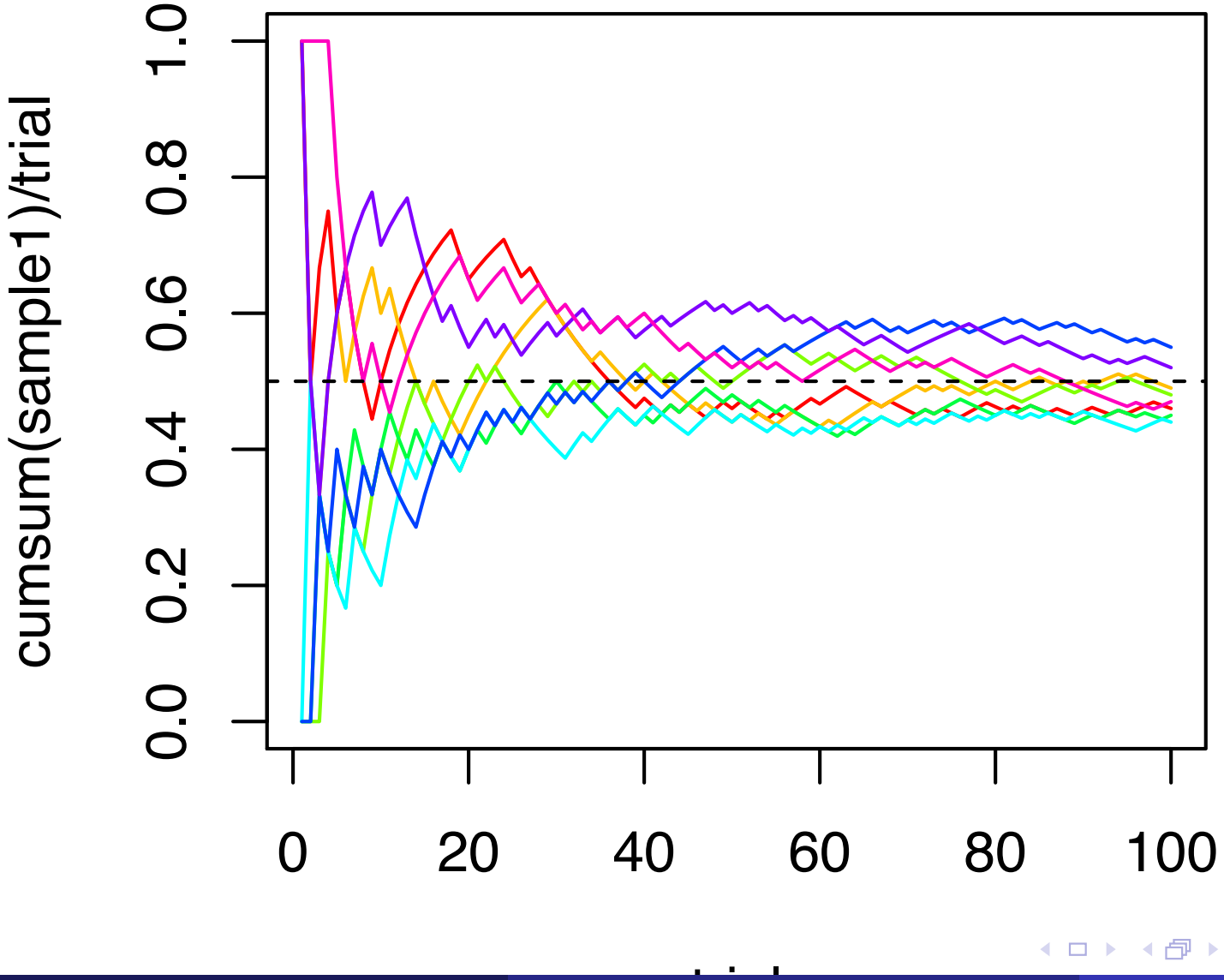


Simulation

```
n=100;

trial=seq(1,100,by=1);

sample1=rbinom(n,1,1/2);
sample2=rbinom(n,1,1/2);
sample3=rbinom(n,1,1/2);
sample4=rbinom(n,1,1/2);
sample5=rbinom(n,1,1/2);
sample6=rbinom(n,1,1/2);
sample7=rbinom(n,1,1/2);
sample8=rbinom(n,1,1/2);

colors=rainbow(8);
```

Simulation

```
plot(trial,cumsum(sample1)/trial,type="l",
col=colors[1],ylim=c(0,1));
lines(trial,cumsum(sample2)/trial,col=colors[2]);
lines(trial,cumsum(sample3)/trial,col=colors[3]);
lines(trial,cumsum(sample4)/trial,col=colors[4]);
lines(trial,cumsum(sample5)/trial,col=colors[5]);
lines(trial,cumsum(sample6)/trial,col=colors[6]);
lines(trial,cumsum(sample7)/trial,col=colors[7]);
lines(trial,cumsum(sample8)/trial,col=colors[8]);
abline(h=0.5,lty=2, col="black");
```


cumsum(sample1)/trial
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100

Empirical Probability

The Law of Large Numbers gives us Empirical Probabilities for events.
Example: Toss a coin. Define the random variable:

$$X = \begin{cases} 1 & \text{heads up} \\ 0 & \text{tails up} \end{cases}$$

Think about sampling n of observations of X, and finding the sample mean $\bar{X}_n$. Let $n \to \infty$, and $\bar{X}_n \xrightarrow{\mathcal{P}} P(\text{heads up})$.